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Educational applications based on diagnostic learning analytics model in the context of big data analytics

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ABSTRACTS

Under the context of big data analytics, Diagnostic Learning Analytics (DLA) offers a novel approach to understanding and improving student learning behaviors. This study proposes a DLA model that integrates the M-Canopy-K-means clustering algorithm with the PSSHRNet-based pose estimation network to perform deep mining and analysis of classroom behavior data. The experimental results show that the proposed M-Canopy-K-means algorithm improves the clustering accuracy by 10 % and 7 % respectively compared with the K-means and Mini-batch K-means algorithms, and achieves an speedup ratio of more than 2 times on small-scale datasets. In behavior recognition, the PSSHRNet-based network achieved lower loss values and faster convergence on MSCOCO and CrowdPose datasets. In practical applications, the model detected that a student with high engagement had an actual participation score of only 2.81, with mobile phone use accounting for 33.29 % of classroom behavior. The study demonstrates the effectiveness of big data-driven DLA models for diagnosing educational behaviors and provides a solid foundation for personalized teaching interventions.

1. Introduction

With the rapid advancement of digitalization and intelligent transformation in education, there is an increasing demand for more precise monitoring, assessment, and optimization of student learning processes. Traditional teaching evaluations, which primarily rely on exam scores, subjective teacher observations, or basic data statistics, fail to comprehensively and dynamically reflect students' behavioral patterns and engagement levels [1]. These limitations often result in the neglect of individual differences and hinder the timely adjustment of instructional strategies, thereby affecting teaching quality and student development [2]. Driven by the progress of information technology and data science, big data analytics has emerged as a vital support tool in educational research [3]. By mining and modeling large-scale, heterogeneous educational data, educators can gain more detailed insights into students' learning needs and behavioral characteristics, thus enabling more scientific and targeted teaching interventions [4]. However, the current applications of big data analytics in education still face several challenges. On one hand, conventional Learning Analytics (LA) methods predominantly rely on static data, such as assignment submissions, transcript records, or platform clickstreams, making it difficult to capture real-time dynamics in students' learning behaviors [5]. On the other hand, commonly used data mining algorithms, such as traditional K-means clustering, are highly sensitive to initial centroid selection and often yield unstable clustering results, limiting their effectiveness in analyzing complex behavioral data. Moreover, traditional behavior recognition methods that depend on handcrafted features exhibit significant performance degradation under variable lighting conditions and occluded classroom environments, leading to inaccuracies in learning status assessment.

To address these challenges, Diagnostic Learning Analytics (DLA) has emerged, focusing on the deep analysis of student behaviors and learning data to diagnose learning obstacles and provide personalized improvement recommendations. Nevertheless, the practical application of DLA remains constrained by two main technical bottlenecks: the lack of efficient and accurate behavioral feature extraction and modeling techniques, and the absence of scalable, high-performance clustering and analytics frameworks capable of handling large-scale educational datasets. Consequently, there is a pressing need to develop a novel DLA model that integrates big data mining and high-precision behavior recognition to overcome these limitations. In response to these gaps, this study proposes a diagnostic learning analytics model that combines big

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data mining with advanced pose estimation techniques. Specifically, the contributions of this study are threefold: (1) proposing the M-Canopy-K-means algorithm based on the MapReduce framework, optimized with the maximum-minimum principle and triangle inequality strategy to enhance clustering accuracy and scalability; (2) designing the PSSHRNet, a high-resolution pose estimation network with integrated spatial and channel attention mechanisms, to achieve precise detection and analysis of student classroom behaviors; (3) conducting extensive experimental validation on large-scale student behavior datasets and public benchmarks (MSCOCO and CrowdPose) to systematically evaluate the model's performance and application potential. This research aims to provide robust data-driven support for personalized education, facilitating intelligent teaching and individualized learning path recommendations, and advancing the development of educational informatization.

2. Related work

In the context of intelligent education, learning analysis (LA) technique has attracted the attention of more scholars. Gumantan A et al. introduced an innovative e-learning approach based on LA technique. The core technique of the methodology was the use of quantitative and descriptive statistical techniques, which offer a wide range of advantages in data processing and interpretation that can help in understanding patterns and trends in the learning process. Its ability to provide detailed and practical feedback can be valuable to both teachers and students [6]. Muñoz J's study provided an in-depth exploration of existing LA approaches and a comprehensive review of the research literature on adaptive learning techniques used in higher education in recent years. The study outlined the importance of adaptive feedback and navigation to help learners process information more effectively and thus improve their learning [7]. Researchers such as Rasmitadila R proposed a LA approach based on situational analysis aimed at providing an in-depth understanding of students' perceptions of the implementation of blended learning approach (BLA) in inclusive education programs. The proposed LA method helped educators and researchers to better understand and improve the BLA approach [8]. Mertha I and Mahfud M successfully developed a history learning assessment methodology which tests students across the board by using the Wordwall web application as an assessment tool. This innovative assessment tool provided a more comprehensive and flexible approach to assessment. This learning assessment tool was evaluated by material experts and proved to be fully feasible [9]. Scholars such as Novalinda R have proposed a LA method based on triangulation techniques, which is widely used in educational research to understand and assess the learning process from multiple perspectives and dimensions. The proposed model was executed in three parts: data reduction, data presentation, and drawing and validation of conclusions. The team's findings revealed that the proposed method can significantly improve personalized learning models with modules [10].

Additionally, Pishtari G et al., addressing the issue of insufficient collaboration between teachers and intelligent systems in existing model-based learning analytics (MbLA), conducted a bibliometric and topic modeling analysis. They revealed that MbLA research tends to focus on individual learning and neglect collaborative learning. They proposed that the transparency of models should be enhanced to foster deeper integration between teachers and AI systems, thereby improving the application of learning analytics in teaching practices [11]. Jiang S et al., addressing the issue of limited application of LA to assess hands-on laboratory skills in K-12 science classrooms, proposed using mobile augmented reality (AR) applications to log process data and combined it with Bayesian networks to mine students' laboratory skills. This approach allows real-time, large-scale measurement of students' experimental skills using LA, and reveals a positive correlation between laboratory skills and conceptual learning [12]. Jin S H et al., addressing challenges students face in applying self-regulated learning (SRL) in online learning environments, proposed supporting SRL in LA through artificial intelligence (AI) applications. They explored students' perceptions of AI applications in supporting SRL. The study revealed that students believed three pedagogical and psychological aspects should be considered for effective utilization of AI applications: learner identity, learner activeness, and learner position [13]. Gao W, addressing the issue of domain-level zero-shot cognitive diagnosis (DZCD) due to the absence of student practice logs in new domains, proposed the LA framework Zero-1-to-3. By pre-training a diagnostic model and introducing dual regularizers, it decouples student states into shared and domain-specific parts, allowing for the transfer and sharing of cognitive signals. This approach addresses the cognitive state propagation goal for new domains and optimizes the cognitive states of cold-start students by generating simulated practice logs, thereby enhancing the accuracy of cognitive diagnosis results [14].

Studies related to the integration of big data and education have been quite popular among researchers in recent years. Cele N's study pioneered the application of big data technology to student data management systems in higher education by proposing a big data early warning system. The system integrated and analyzed a variety of data in order to gain a deeper understanding of students' socio-economic and academic needs and to develop more effective intervention strategies accordingly. Through this big data early warning system, schools can quickly identify students who may be facing academic difficulties and provide timely assistance to effectively improve student learning outcomes [15]. Researchers such as Fischer C have proposed a novel data-driven approach to support informed decision making and improve educational efficiency. This approach leverages the power of big data and artificial intelligence to deeply analyze and interpret large amounts of diverse educational data to gain useful knowledge and insights. This data-driven approach not only helps educators understand students' learning behaviors, preferences, and needs, but also predicts future learning trends and outcomes [16]. Scholars such as Gao P have conducted an in-depth study on distance education in the context of big data, especially exploring the effectiveness of next-generation mobile networks and IT in education. The study focused mainly on how to utilize advanced technological tools and methods in order to improve the efficiency and effectiveness of education. The results showed that the application of next-generation mobile networks and IT in education can significantly improve the efficiency of teaching and enhance students' learning experience [17]. The study by Khan S and Alqahtani S delved into how big data can be utilized in the field of education and provided a comprehensive review of the research literature on big data in education for the period 2010 to 2020. The study was application-based and explored how big data can be utilized to improve the educational process. In addition, the study used two different methods to validate the educational process, including quantitative and qualitative analysis to ensure the accuracy and reliability of the findings [18]. Berendt B delved into the benefits and risks of AI in education, especially in relation to basic human rights. The study examined in detail the potential of AI and "big data" in the education system, especially in terms of real-time monitoring and improving the efficiency of education. These technologies can also help educators to quickly identify and solve teaching and learning problems, contributing to the improvement of educational efficiency [19].

In summary, despite significant progress in the application of big data and LA in the education field, current research still faces limitations in real-time monitoring of student learning behaviors and providing personalized teaching interventions. Existing methods mainly rely on static learning outcomes and post-hoc data, making it difficult to diagnose and provide feedback on students' dynamic learning processes and behavioral changes in a timely manner. Particularly in the areas of large-scale educational data processing, precise assessment of learning states, and multi-dimensional personalized support, technical and methodological challenges remain. To address these challenges, this paper proposes an innovative DLA model, combining big data mining and student classroom behavior recognition, aimed at providing solutions for personalized teaching and real-time learning monitoring in educational practice. The model utilizes the M-Canopy-K-means clustering algorithm based on MapReduce and the high-resolution PSSHRNet pose estimation network to capture students' behavioral changes in real time, build behavioral models, and diagnose cognitive states. Unlike existing learning analytics methods, the DLA model not only focuses on the real-time collection and analysis of behavioral data but also integrates multi-dimensional information such as cognitive, emotional, and behavioral data, creating a complete feedback loop from data collection to teaching intervention, providing more precise personalized teaching support.

The innovation of this study lies in the introduction of a cross-domain, real-time, and detailed learning analytics method, offering a new framework for deep collaboration between teachers and technology in intelligent education systems. Compared to traditional methods, the DLA model allows for more accurate and dynamic assessments of students' learning states and provides personalized interventions based on real-time data, significantly enhancing the effectiveness of learning analytics and improving teaching quality.

3. Diagnostic learning analytics model based on big data technology

The section begins with a detailed description of the big data mining techniques in the DLA model, discussing their important role in processing massive learning data and mining students' learning behavior patterns. Based on this, an innovative DLA model is proposed, which focuses on students' behaviors in the classroom, and through the in-depth analysis and understanding of these behaviors, aims to diagnose students' learning problems and provide them with personalized learning suggestions.

3.1. Model construction of learning analytics based on big data technology

LA is an emerging approach to learning science that focuses on analyzing and understanding learners' behaviors and performances in the learning process through systematic technical means [20–21]. The main goal of LA is to discover the characteristics of learners' learning behaviors as well as learning style characteristics, which include how they approach new knowledge, how they process information, and their coping strategies in the face of difficulties and challenges [22]. This analysis can be carried out in a number of ways, for example, by examining the learner's activities in the Learning Management System, including the pages he/she visits, the tasks he/she completes, the statements he/she makes on the discussion boards, and so on. In addition, learners' performance and feedback can be used to further understand their learning. The results of LA can help educators and learners themselves to adjust their learning strategies. If the analysis shows that a learner is struggling in a particular area, then they can be assisted through individualized instruction and tutoring [23]. In addition, if the analysis shows that a learner excels in a domain, then deeper learning resources can be provided to meet their learning needs. The general LA system architecture is shown in Figs. 1 and 2.

Intelligent digital education services provide personalized education solutions with the help of LA for stakeholders such as students, teachers, and educational decision makers. As the data-driven nature of business comes to the fore, the need for data analytics will continue to grow in the future [24]. In this process, analysis algorithms become key, especially

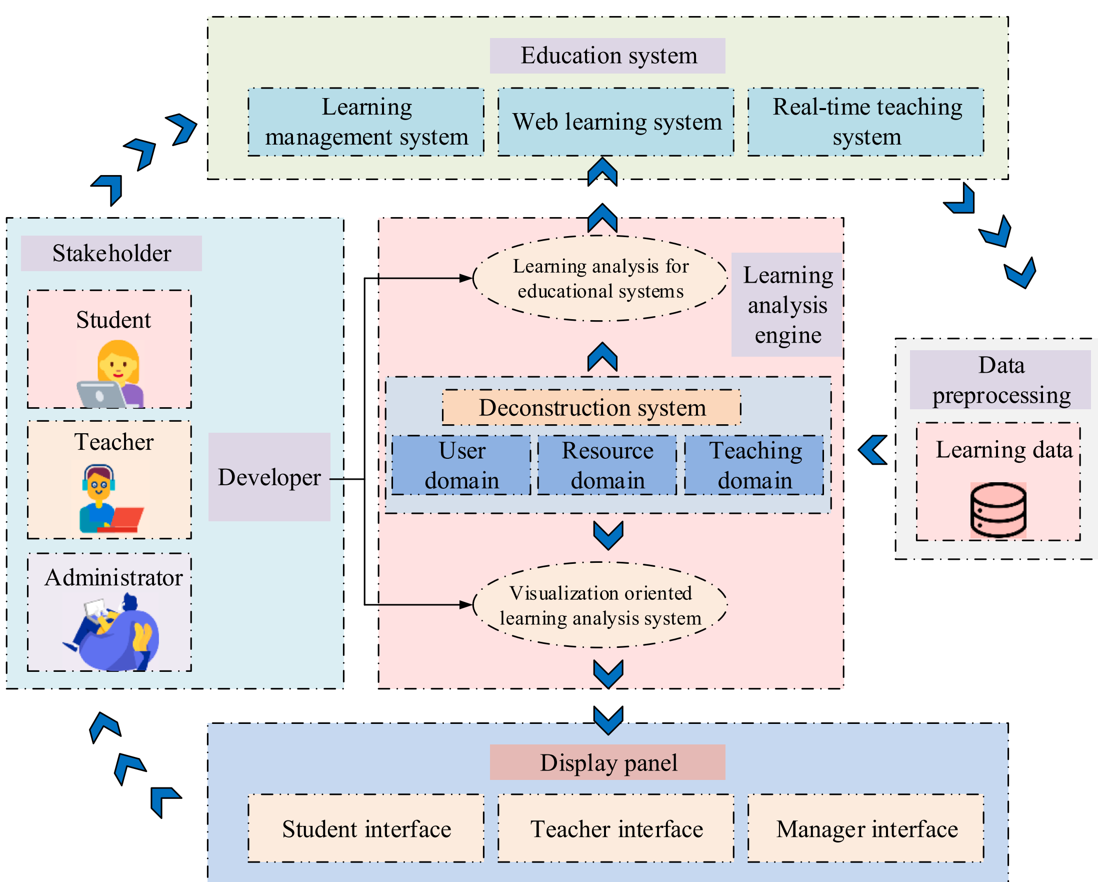

Fig. 1. Learning analysis system architecture.

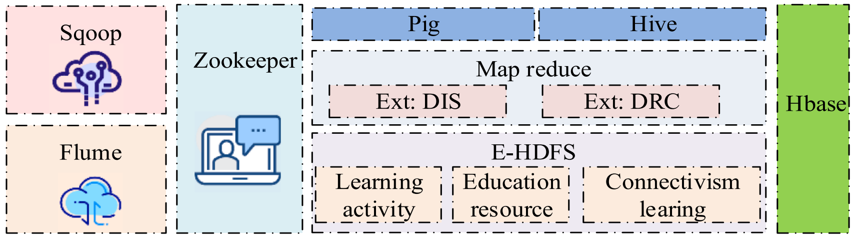

Fig. 2. Learning analysis distributed computing system architecture based on Hadoop.

the need to build algorithmic modules that can be scaled on-demand and to design parallel algorithms for big data in order to achieve more accurate LA and to promote the intelligent and digital process of educational services. In intelligent digital education services, data are collected through LA data aggregation system and are acquired and processed by LA distributed computing system based on Hadoop. Hadoop is a distributed storage system that builds a complete big data ecosystem, including utilities, distributed file systems, data analysis and storage platforms, and application layers. Its core computing model MapReduce supports distributed processing of large datasets, realizes horizontal and linear scaling, and realizes parallel processing through two steps of Map and Reduce to improve processing efficiency and speed.

The K-means algorithm, as a representative clustering algorithm, has a fast convergence rate, but suffers from the problems of randomness in cluster centroid selection and the need to predetermine the number of clusters. To improve these shortcomings, the study constructed the Canopy-K-means algorithm (M-Canopy-K-means) combined with MapReduce distributed framework to improve the model accuracy and scalability [25]. However, the Canopy algorithm still has the problem that the initial threshold is artificially specified. Therefore, in order to optimize the selection of Canopy centroids, the "max-min principle" is introduced to further improve the accuracy and usability of the clustering algorithm. In the centroid selection method based on the "Max--Min Principle", in order to make the distance between canopy centroids as large as possible, the $n+1$th canopy centroid is selected as the largest of the shortest distances from the previous n centroids. This method optimizes the data set division and improves the accuracy and efficiency of clustering, and its calculation formula is shown in Eq. (1).

$$\begin{cases} GistList = \min\{d_1, d_2, d_3, \cdots, d_n\} \\ DistMin(n+1) = \max\{DistList\} \end{cases} \tag{1}$$

In Eq. (1), d_n denotes the minimum distance between the nth centroid and the candidate data point, and $DistList$ denotes the set of minimum distances between the first n centroids and the candidate data points. $DistMin(n+1)$ denotes the maximum value of the minimum distance in the set $DistList$. Based on the "Max-Min Principle", when the number of centroids is close to the optimal number, the variation of $DistMin(n+1)$ is the largest. In order to determine the optimal number of Canopy centroids and the region radius $Depth(i)$, a depth indicator L is introduced to describe the magnitude of change of centroids.

$$\begin{aligned} Depth(i) = &|DistMin(i) - DistMin(i-1)| \\ &+|DistMin(i) - DistMin(i-1)| \end{aligned} \tag{2}$$

In Eq. (2), $Depth(i)$ achieves the maximum value when i is close to the true number of clusters, and $T_1 = DistMin(i)$ is set at this time to make the clustering result optimal. In order to reduce the computational volume of the K-means algorithm and accelerate the convergence speed, this study adopts the triangular inequality theorem to optimize this algorithm and reduce the unnecessary distance computation in the process of data object division. For any vector x, b, c in the Euclidean space, Eq. (3) is satisfied.

$$\begin{cases} d(x,b) + d(b,c) \geq d(x,c) \\ d(x,b) - d(b,c) \leq d(x,c) \end{cases} \tag{3}$$

Assume that x_p is any vector in the dataset and c_i is the current cluster center of vector x_p. x_p is known and x_p is any cluster center except x_p, if x_p, then the expression is showed in Eq. (4).

$$d(x_p, c_i) \leq d(x_p, c_j) \tag{4}$$

The improvement method based on triangular inequality can effectively reduce the redundant distance calculation of K-means. This study mainly improves the Canopy-K-means algorithm from two aspects, firstly, introducing the "maximum-minimum principle" to optimize the selection of Canopy centroids; and then optimizing the K-means algorithm by using triangular inequalities to reduce the redundant distance calculations and accelerate the convergence speed of the algorithm. The improved algorithm is mainly divided into two stages, and its flow is shown in Fig. 3.

M-Canopy-K-means, mentioned in this research, plays a key role in educational data mining, with its ability to handle large-scale educational data, such as student learning behavior and performance. Parallel computing capabilities allow faster data analysis, including clustering, classification, and association rule mining. In addition, MapReduce-based machine learning algorithms can predict student learning effectiveness and identify and intervene in learning difficulties in a timely manner, providing educational researchers with a tool to effectively process and analyze large-scale educational data. The application of big data mining techniques in the diagnosis of students' classroom behavior has unique features and advantages.

3.2. Diagnostic learning analytics model construction incorporating M-Canopy-K-means and PSSHRNet

The DLA model is based on big data mining and machine learning technologies, which conduct in-depth analysis and diagnosis of students' classroom behaviors to optimize teaching methods and improve student learning outcomes. Traditional learning analytics methods primarily rely on static data and performance assessments, while the DLA model collects real-time classroom behavior data, such as attendance, engagement, and participation in discussions, to help teachers understand students' learning attitudes, habits, and potential learning difficulties [26]. The core innovation of this model lies in utilizing machine learning algorithms to deeply mine and analyze these behavioral data, enabling the early identification of possible issues and providing timely intervention, thereby enhancing the effectiveness and targeting of education.

This study combines the M-Canopy-K-means clustering algorithm with machine learning methods to construct the DLA model. By real-time monitoring and analyzing students' classroom behaviors, this model provides accurate learning state diagnosis for educators and allows personalized learning plans to be developed based on students' learning progress and behavioral patterns. Moreover, students' classroom behaviors are closely linked with pose estimation, which further improves the diagnostic accuracy and real-time capability of the model [27]. Posture estimation identifies and analyzes students' body postures,

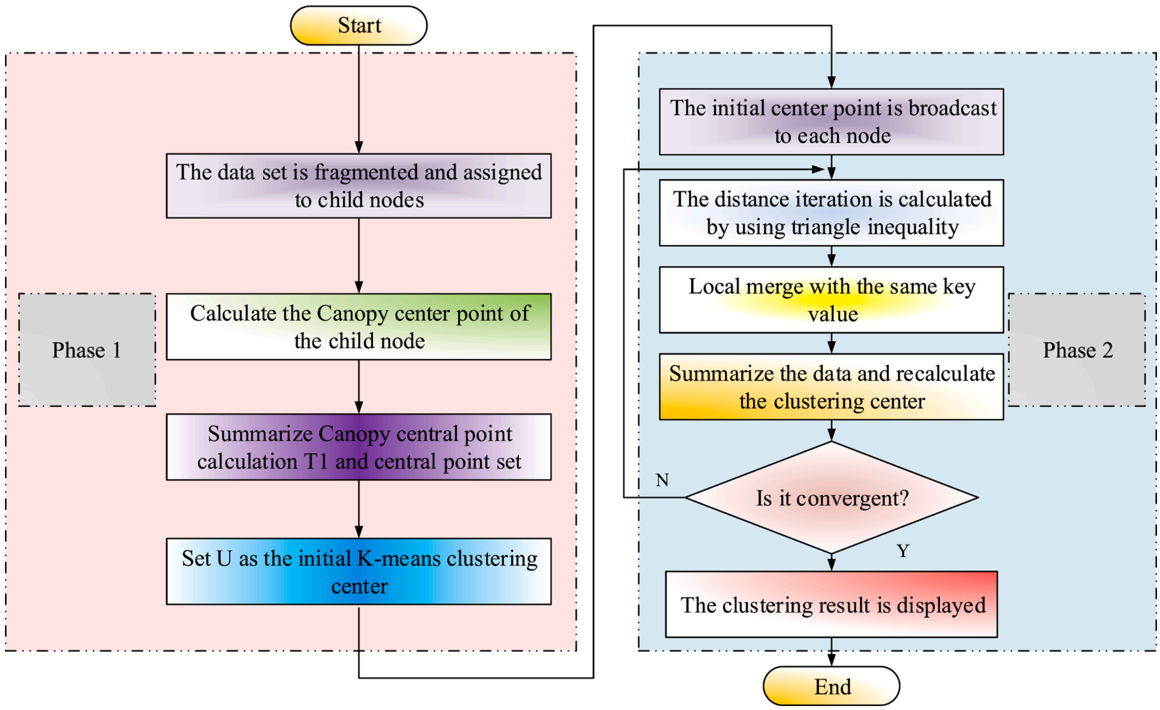

Fig. 3. Operation flow of Canopy-K-means.

reflecting their level of classroom engagement and attention. Therefore, posture estimation provides teachers with an important teaching basis as a quantitative criterion of students' classroom behavior. To accurately estimate students' classroom postures, the study proposes a high-resolution PSSHRNet network based on spatial and channel AM. This network introduces a ResNet module with parallel spatial and channel attention on top of HRNet for extracting high-resolution representations with contextual relationships to improve the quality of features extracted by the model.

Deep neural networks face two major problems: gradient explosion/vanishing and performance degradation. The gradient problem can be solved by adding regularization and activation layers, while the constant mapping of ResNet can alleviate the performance degradation [28]. Since it is challenging to learn the constant mapping directly, the fitted residual function is introduced in order to construct the constant mapping in deep networks as shown in Eq. (5).

$$F(x) = H(x) - x \tag{5}$$

In Eq. (5), x denotes the input to this residual structure, $F(x)$ denotes the residual function that needs to be learned to be fitted by the solver, and $H(x)$ denotes the mapping formula. The advantages of ResNet networks include sensitive residual functions, constant connections across layers without additional parameters, reduced optimization difficulty, and a structure that supports backward and forward propagation. Although ResNet effectively solves the deep network problem, it has not overcome the noise in feature extraction [29]. By adding AM to ResNet, key features can be strengthened, redundant features can be suppressed, and adaptive learning can be realized. The expression of ResNet function by adding AM is shown in Eq. (6).

$$y = \mathrm{Relu}(F(x) + A(x) + x) \tag{6}$$

In Eq. (6), y denotes the output of the ResNet network with introduced AM and $F(x)$ denotes the residual function. $A(x)$ denotes the introduced AM and x denotes the input of the ResNet network with AM. AM contributes to pixel-level human pose estimation by establishing remote connections in the channel and spatial information and calculating inter-pixel dependencies [30]. The study proposes a pixel-level based AM that adds a rotation operation for cross-dimensional interaction and maintains high-resolution features to capture fine-grained features, as shown in Fig. 4.

Given the input high resolution representation $X \in R^{C \times H \times W}$, the high resolution output after AM can be expressed as Eq. (7).

$$Z(X) = u(X) + v(X) \in R^{C \times H \times W} \tag{7}$$

In Eq. (7), $u(X) \in R^{C \times H \times W}$ denotes the feature map after introducing the interaction between the channel dimension and the height or width dimension in the channel attention module, and $v(X) \in R^{C \times H \times W}$ denotes the output of the spatial attention module. The channel attention module is the first two branches in the PSSNet structure. The first branch computes the interaction information of (C, H) as shown in Eq. (8).

$$X_H = R(X) \tag{8}$$

In Eq. (8), R denotes the counterclockwise rotation operation along the fixed dimension. The second branch computes the interaction information of (C, W) as shown in Eq. (9).

$$X_W = R(X) \tag{9}$$

The spatial attention module is the third branch in the structure of PSSNet. This branch computes the interaction information between the (H, W) dimensions of spatial information. For a given high-input high-resolution representation $X \in R^{C \times H \times W}$, the dimensions of the input resolution representation are first converted to $C/2 \times H \times W$ by a convolution operation, after which the former feature map is sequentially subjected to global pooling, pooling, and Soflmax operations to obtain a vector of size $1 \times C/2$. At the same time, the other branch obtains the tensor of size $C/2 \times HWB$ by a reshaping operation as shown in Eq. (10).

$$A^{sp}(X) = F_{RS}\left[\delta_3\left(F_{SF}\left(\delta_1\left(F_{GP}\left(W_q(X)\right)\right)\right) \times \delta_2(W_v(X))\right)\right] \tag{10}$$

In Eq. (10), W_q and W_v denote the convolutional operations with convolutional kernel size 1×1. σ denotes the reshaping operation and F_{RS} denotes the reshaping and Sigmoid operation. F_{RS} denotes the dot product operation of vectors. F_{GP} denotes the Softmax operation whose functional expression is shown in Eq. (11).

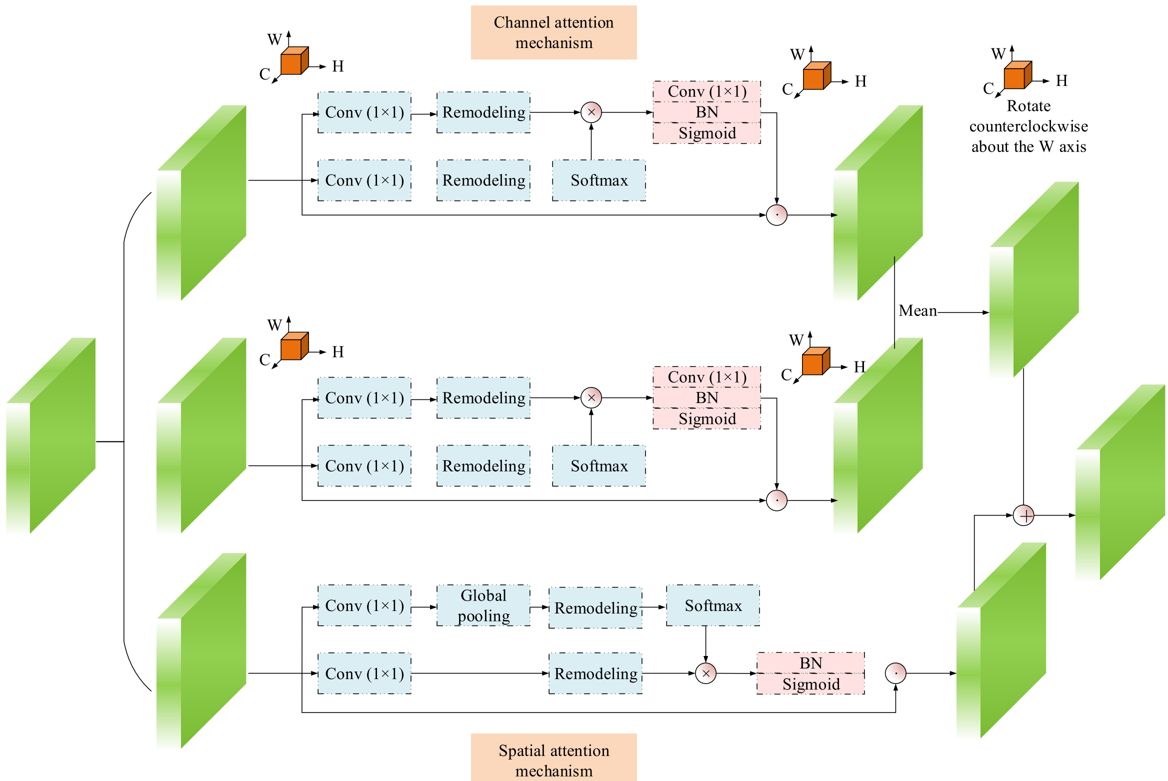

Fig. 4. PSSNet structure.

$$F_{GP} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X(:, i, j) \tag{11}$$

After this, element-by-element multiplication with the input high-resolution feature map weights the original input to obtain a spatial weight-based high-resolution feature map, as shown in Eq. (12).

$$Z^{sp} = A^{sp}(X) \otimes X \tag{12}$$

In Eq. (12), W_q denotes the high-resolution feature map and $\otimes$ denotes the spatial dimension of the multiplication operation. Thus, given the input high-resolution representation $X \in R^{C \times H \times W}$, the process of obtaining the fine pixel-level attention representation Z from PSSNet AM is shown in Eq. (13).

$$Z = Z^{sp}(X) + Z^{ch}(X) \tag{13}$$

In Eq. (13), Z^{sp} denotes the high-resolution feature map based on spatial weights, and Z^{ch} denotes the feature map after weighted channels and height interactions in the spatial domain. The PSSHRNet network in the study uses HRNet as the backbone to recover high-resolution heat maps using the anti-convolution structure of HigherHRNet. In the feature extraction stage, ResNet is replaced with PSSNet to enable PSSHRNet to capture fine pixel-level features, which are essential for high-precision implementation of keypoint localization, detection, and classification. The PSSHRNet network structure is shown in Fig. 5.

This research combines M-Canopy-K-means with PSSHRNet to construct a new DLA model. This model not only inherits the advantages

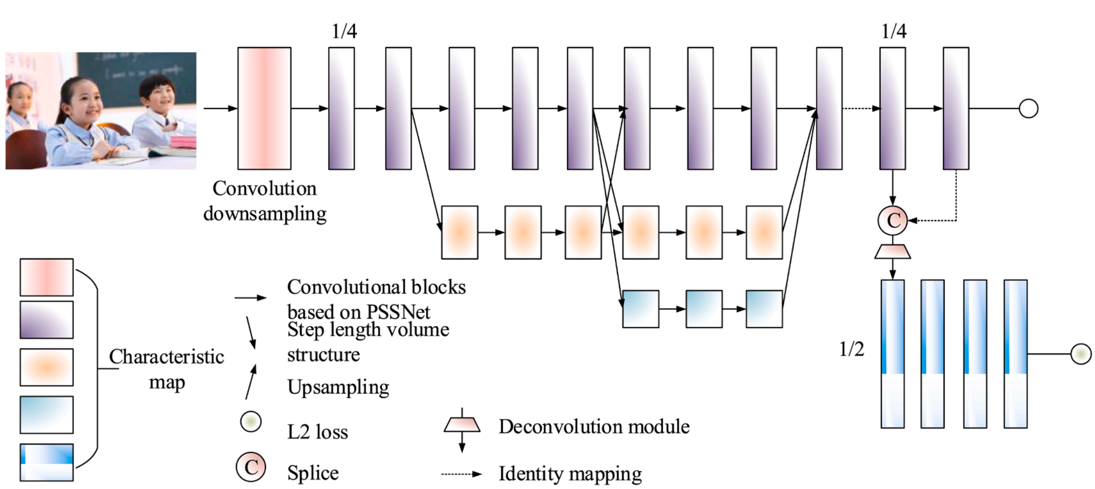

Fig. 5. PSSHRNet network structure.

of M-Canopy-K-means in data clustering, but also takes advantage of the powerful ability of machine learning in data processing and analysis, thus providing a new and effective means for deeper and more accurate analysis and diagnosis of students' classroom behaviors, and the constructed model is shown in Fig. 6.

4. Effectiveness of diagnostic learning analytics modeling in real-world applications

This section begins with a detailed validation of the data mining effectiveness of the proposed model. The performance of the model is comprehensively evaluated through experimental data and results, including key indicators such as clustering effect, processing speed, accuracy, etc., thus proving the superior performance of the model in data mining. Then, the effectiveness of the model in practical applications is further explored.

4.1. Performance validation of M-Canopy-K-means

For this research, a Hadoop cluster was built on a server equipped with 17 CPU, 16 G RAM, 2 TB hard disk, including 1 Master and 5 Slave nodes. Each node is equipped with 2GB RAM, 200 G hard disk, running CentOs 6.5, jdk1.8.0–181, and Hadoop 2.7.3. All algorithms are done in Java, and the development tool is Eclipse. the experimental datasets are preprocessed microblog text with 100 M, 500 M, 1 G, and 2 G of data, which are used to test M- Canopy-K-means algorithm.

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In Fig. 8(a), M-Canopy-K-means is significantly more efficient than the other two algorithms for the same number of nodes, and the M-Canopy-K-means algorithm optimizes the selection of centroids through the "Max-Min Principle" to reduce the number of iterations, and utilizes the Triangular Inequality Theorem to reduce the redundant distance calculations, thus significantly improving the execution efficiency relative to the other algorithms. The M-Canopy-K-means algorithm reduces the number of iterations by optimizing the centroid selection through the "maximum-minimum principle", and reduces the redundant distance calculation by using the triangular inequality theorem, which significantly improves the efficiency of the algorithm in comparison with the others. In Fig. 8(b), in order to verify the parallel efficiency of M-Canopy-K-means on datasets of different sizes, four datasets, namely, 100 M, 500 M, 1 G, and 2 G, are clustered on 1, 3, and 5-node Hadoop clusters, and the average computation time and speedup ratio are calculated. The results show that for small datasets, the acceleration ratio of M-Canopy-K-means is smooth when the cluster node is 3, and after that it decreases slightly with the increase of nodes, implying a waste of resources. For large datasets, the acceleration ratio rises with the increase of nodes, but the growth decreases gradually. This proves that M-Canopy-K-means can effectively improve the clustering efficiency when executed in parallel on large-scale data.

Fig. 9 shows the clustering results of K-means, Mini-batch K-means and M-Canopy-K-means. In the figure, the clustering results of M-Canopy-K-means are better fitted to the true values than the other two. Although K-means and Mini-batch K-means also have their unique advantages when dealing with data clustering, they still have more errors.

4.2. Performance validation of diagnostic learning analytics models incorporating M-Canopy-K-means and PSSHRNet

To evaluate the performance of the proposed DLA model, two publicly available datasets, MSCOCO and CrowdPose, were selected for experiments in this study. MSCOCO is a large image dataset widely used for training and validating computer vision models, and includes a wide variety of scenes, objects, and people. CrowdPose, on the other hand, is a dataset specialized in crowd pose estimation, whose complex crowd scenes and complex pose challenges make it ideal for validating model performance. Utilizing these two datasets enables comprehensive and effective validation of the proposed model performance in diverse scenes and complexities.

Fig. 10 shows the loss function curves for the model training process on the MSCOCO and CrowdPose datasets. In each group of images in the figure, the dark green solid line represents the average changing trend of the model loss during the training process, and the light green area

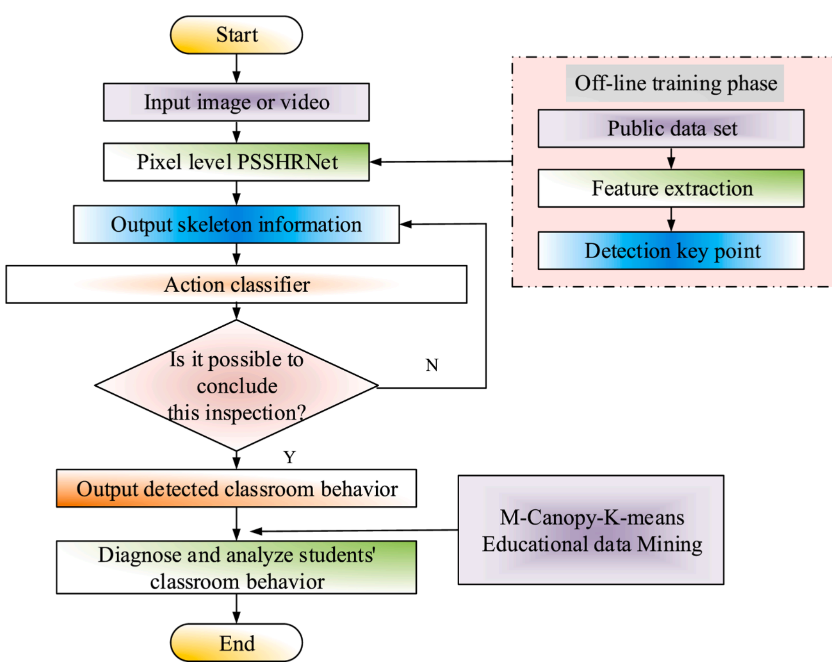
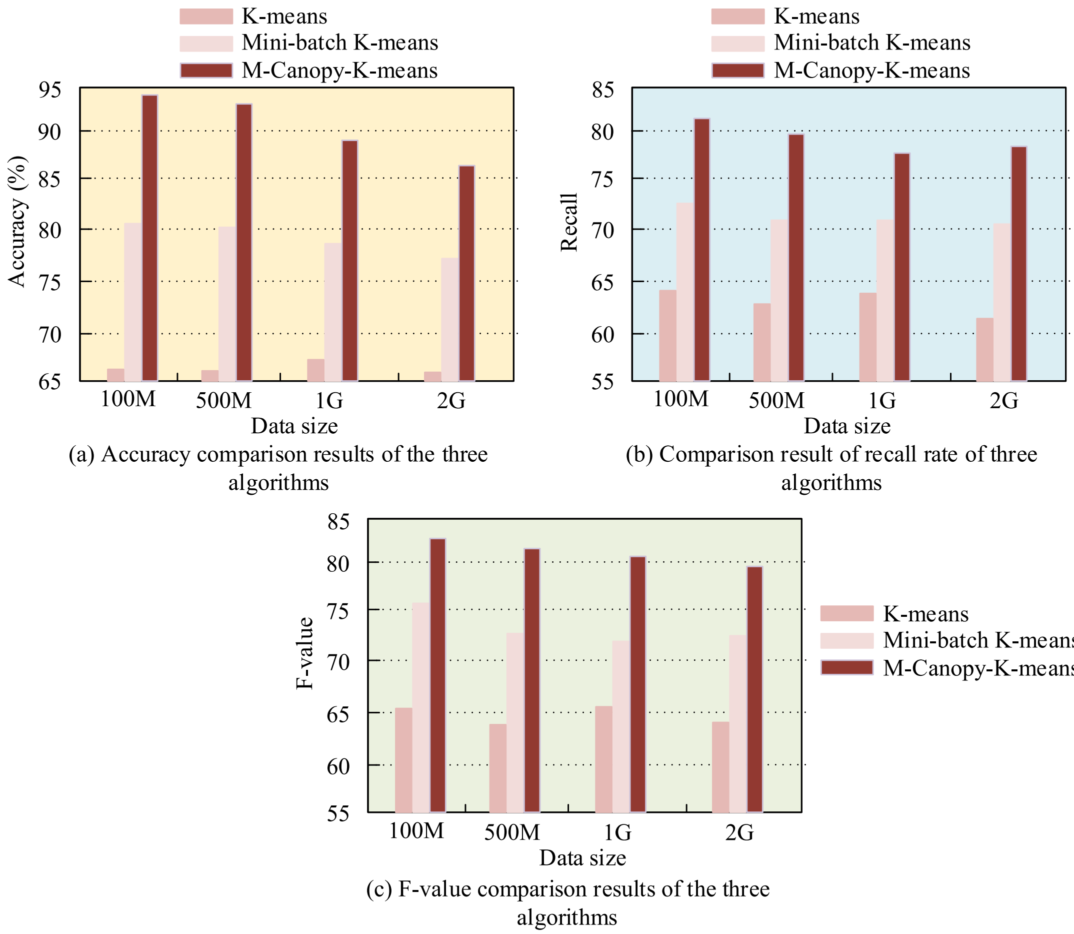


Fig. 6. Diagnostic learning analytics model combining M-Canopy-K-means and PSSHRNet.

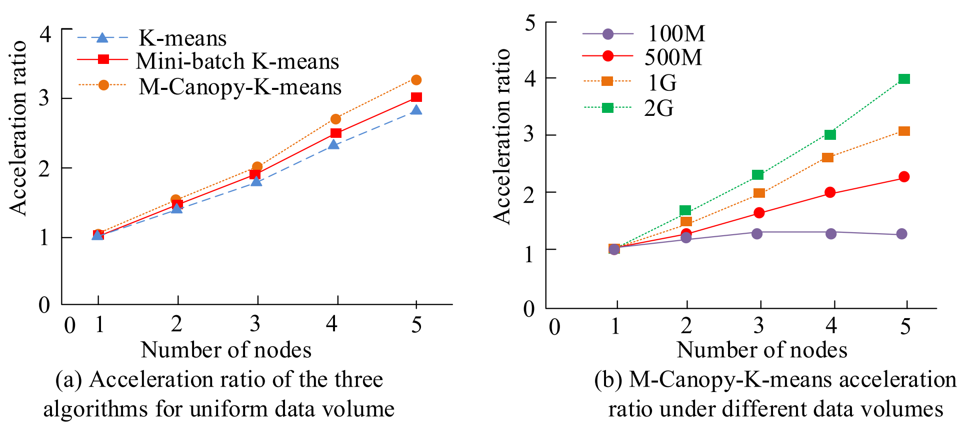


(a) Accuracy comparison results of the three algorithms

(b) Comparison result of recall rate of three algorithms

(c) F-value comparison results of the three algorithms

Fig. 7. Comparison results of accuracy, recall rate and F-number of the three algorithms.



(a) Acceleration ratio of the three algorithms for uniform data volume

(b) M-Canopy-K-means acceleration ratio under different data volumes

Fig. 8. Scalability analysis results of the algorithm.

represents the fluctuation range of the loss value (such as standard deviation or confidence interval). Both curves show that the loss drops sharply and then smoothes out, indicating that the network parameters have reached an appropriate state. Compared with MSCOCO, the loss of CrowdPose is smaller but more oscillatory, probably due to the fact that CrowdPose has a more complex occlusion scene and its data volume is smaller than MSCOCO.

The proposed modeling system uses time as a cue to carry out the analysis of students' classroom learning behaviors. The model analyzes students' classroom learning behavioral engagement through time cues. Fig. 11(a) and (b) show that behavioral engagement remains steady during the course and decreases before the end of the class. Learning disengagement declined early in the lesson and gradually increased towards the end. In Fig. 11(c), "looking at the blackboard" and "looking at the computer" are the main behaviors, presuming that the teacher adopts a lecture approach; "communicating" and "raising hands to answer questions" have peaks, suggesting that the teacher implements effective communication and discussion.

The model analyzed individual student behavioral inputs. Fig. 12 characterizes the behavioral distribution of a student with high disengagement in the classroom. In Fig. 12(a), the student's classroom engagement is only 2.81, and the frequency of cell phone looking

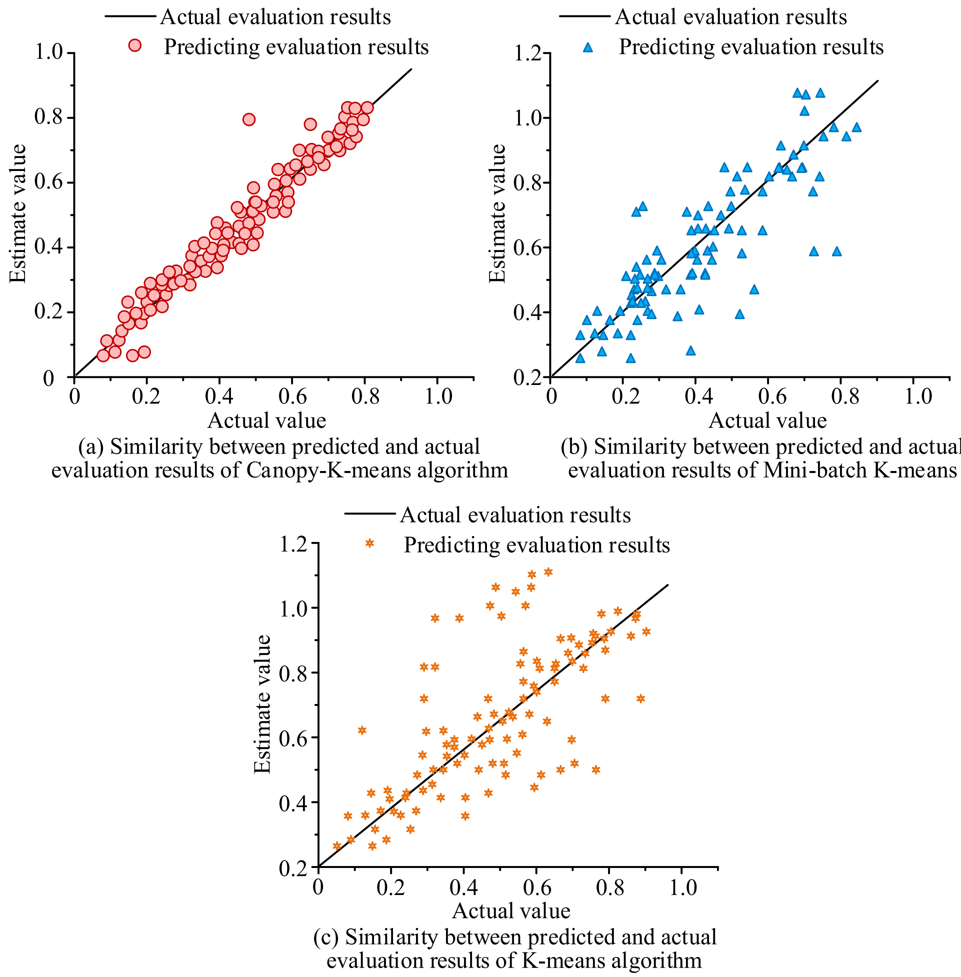


Fig. 9. Clustering effect of the three algorithms.

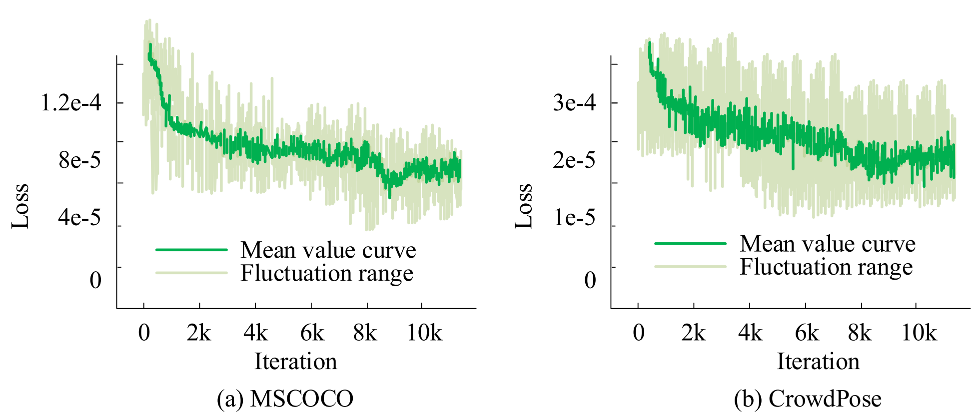
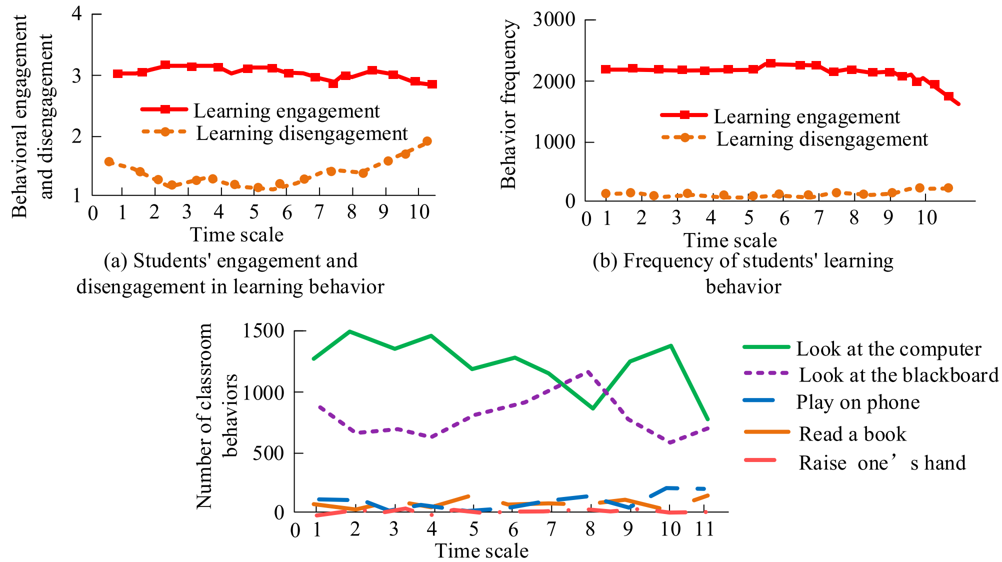


Fig. 10. Changes of the model's loss function on the two data sets.

behavior is 33.29 % of the total classroom behavior. In Fig. 12(b), it reveals that the student showed significant disengagement by looking at his cell phone less at the beginning of the course but almost the entire time later on.

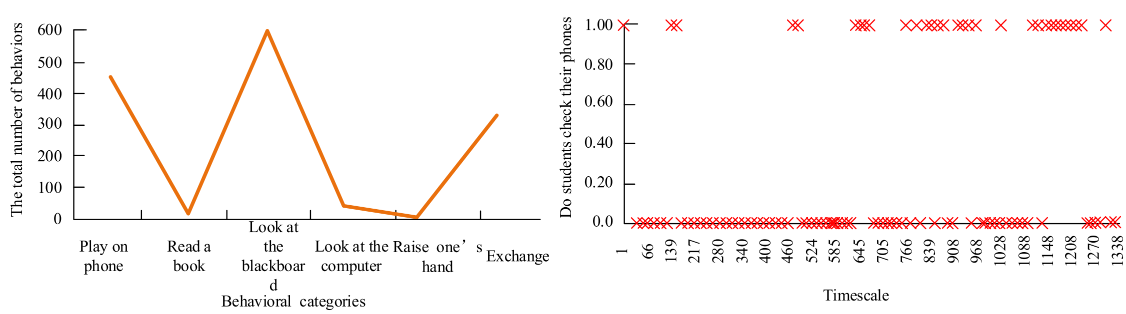
5. Conclusion

In the context of big data analytics, the application of DLA technology in the education field has gained increasing attention. This study proposes a DLA model that integrates M-Canopy-K-means and PSSHRNet, aimed at deeply analyzing individual student behaviors to improve educational quality and teaching effectiveness. Experimental results indicate that M-Canopy-K-means improves accuracy by approximately 10 % and 7 % compared to K-means and Mini-batch K-means, respectively. Further analysis shows that M-Canopy-K-means achieves a stable acceleration ratio for small datasets when the number of clustering nodes is 3, with a slight decrease as the number of nodes increases. This demonstrates the model's good scalability for small datasets. In addition, M-Canopy-K-means outperforms traditional K-means and Mini-batch K-means algorithms in clustering accuracy and



(a) Students' engagement and disengagement in learning behavior
(b) Frequency of students' learning behavior
(c) The frequency dynamics of individual behavioral actions over time

Fig. 11. Analysis results of students’ classroom learning behavior.



(a) The distribution of classroom behavior of a certain student
(b) Distribution of mobile phone viewing behavior among a certain student

Fig. 12. Tracking results of students’ individual engagement in learning behavior.

computational efficiency, showcasing its broad application potential in educational data analysis.In practical application, the DLA model reveals that student classroom engagement is relatively low, with a score of 2.81, and mobile phone usage accounts for 33.29 % of total classroom behavior. These results provide valuable data for educators, helping them gain deeper insights into student behavioral patterns and offering support for personalized teaching interventions. Through precise behavioral data analysis, educators can more effectively identify learning obstacles and needs. This study not only highlights the importance of DLA technology in educational applications but also provides practical tools for educators to analyze student behaviors. However, the study has some limitations, particularly that the proposed model does not fully account for other factors influencing student behavior, such as personal characteristics, family background, and other social factors. Future research should further explore the impact of these factors on student behavior and incorporate them into learning analytics models to enhance the adaptability and accuracy of the models.

6. Future research directions

As this study has demonstrated the potential of the DLA model in analyzing student behaviors and providing personalized educational interventions, several avenues remain for further exploration. To fully realize the potential of DLA in diverse educational settings, future research could focus on the following areas:

1. Incorporating Personalized Factors: Future research could consider including personal characteristics, family background, and other socio-cultural factors into the DLA model. These elements could provide a more comprehensive understanding of student learning behaviors, thus enhancing the model’s ability to offer personalized and context-aware interventions.
2. Real-time Data Integration and Feedback Mechanisms: As DLA models evolve, integrating additional types of real-time data—such as physiological data, emotional states, and contextual factors—along with behavioral data, will allow for more dynamic and precise assessments of student learning. Building intelligent feedback systems to adjust learning paths in real-time based on ongoing data will further optimize educational outcomes and support adaptive learning.
3. Cross-disciplinary Applications: Although this study primarily focuses on classroom behavior recognition and student learning assessment, the applicability of the DLA model can be expanded to different disciplines and various educational environments, including online learning and blended learning. Future work could explore the challenges and opportunities of applying DLA in these diverse contexts.
4. Algorithm Optimization and Big Data Applications: Future studies could optimize the M-Canopy-K-means clustering algorithm and integrate it with other advanced machine learning methods to improve the efficiency of handling large-scale educational datasets.

Additionally, exploring the potential of techniques like transfer learning could provide valuable insights into how DLA models can be adapted across different domains and educational systems.

CRediT authorship contribution statement

Ruxia Jiang: Writing – original draft, Validation, Visualization. **Qun Liu:** Writing – review & editing, Validation. **Jianhong Ren:** Writing – review & editing, Validation.

Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose.

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